

- Description

This module manages a smart drone delivery system, ensuring that:

1. The system continuously monitors the drone's current payload (in kilograms) to ensure it does not exceed the maximum payload capacity.
2. If the payload exceeds the maximum capacity, an overload warning is triggered and the drone will not be cleared for flight.
3. The drone's flight speed is automatically adjusted based on the payload: as payload increases, the flight speed is reduced to conserve energy and maintain safety.

This system helps optimize drone delivery efficiency, maintain flight safety, and ensure energy-efficient operation.

- Specification

**module** SmartDronePayloadControl

**var** currentPayload, maxPayload, baseFlightSpeed, currentFlightSpeed: real

**process** updatePayload(newPayload: real) returns (currentPayload: real, currentFlightSpeed: real)

**functional scenario form:**

currentPayload ≤ maxPayload && currentFlightSpeed = baseFlightSpeed \* (1 - 0.5 \* (currentPayload / maxPayload)), Flight speed is adjusted according to payload.

||

currentPayload > maxPayload && currentFlightSpeed = 0, Overload warning triggered, flight speed set to 0.

**end process**

**end module**

- Java Code

```
package code;
public class c86_SmartDroneDelivery {
    private double maxPayload;
    private double currentPayload;
    private double baseFlightSpeed;
    private double currentFlightSpeed;
    public c86_SmartDroneDelivery(double maxPayload, double baseFlightSpeed, double initialPayload) {
        this.maxPayload = maxPayload;
        this.baseFlightSpeed = baseFlightSpeed;
        this.currentPayload = initialPayload;
        this.currentFlightSpeed = baseFlightSpeed;
        adjustFlightSpeed();
    }
    public void updatePayload(double newPayload) {
        currentPayload = newPayload;
    }
}
```

```

        if (currentPayload > maxPayload) {
            System.out.println("WARNING: Payload exceeds maximum capacity! Flight clearance denied.");
            currentFlightSpeed = 0;
        } else {
            adjustFlightSpeed();
            System.out.printf("Payload updated to: %.2f kg, Adjusted Flight Speed: %.2f km/h\n", currentPayload, currentFlightSpeed);
        }
    }

    private void adjustFlightSpeed() {
        if (currentPayload <= maxPayload) {
            double loadFactor = currentPayload / maxPayload;
            currentFlightSpeed = baseFlightSpeed * (1 - 0.5 * loadFactor);
        }
    }

    public double getCurrentFlightSpeed() {
        return currentFlightSpeed;
    }

    public double getCurrentPayload() {
        return currentPayload;
    }

    public static void main(String[] args) {
        c86_SmartDroneDelivery drone = new c86_SmartDroneDelivery(10.0, 60.0, 3.0);

        // testcase-VT:
        // drone.updatePayload(-55.0);
        // drone.updatePayload(-71.9);
        // drone.updatePayload(-41.7);
        // drone.updatePayload(81.3);
        // drone.updatePayload(-37.2);
        // drone.updatePayload(94.2);
        // drone.updatePayload(-59.9);
        // drone.updatePayload(54.6);
        // drone.updatePayload(77.8);
        // drone.updatePayload(-78.5);
        // drone.updatePayload(27.1);
        // drone.updatePayload(37.1);
        // drone.updatePayload(32.6);
        // drone.updatePayload(-45.0);
        // drone.updatePayload(-54.5);
        // drone.updatePayload(-50.6);
    }

```

```
// drone.updatePayload(1.2);
// drone.updatePayload(-58.2);
// drone.updatePayload(25.4);
// drone.updatePayload(39.7);
// drone.updatePayload(85.6);
// drone.updatePayload(-51.2);
// drone.updatePayload(-94.3);
// drone.updatePayload(-56.5);
// drone.updatePayload(-42.8);
// drone.updatePayload(20.5);
// drone.updatePayload(41.8);
// drone.updatePayload(30.4);
// drone.updatePayload(-66.9);
// drone.updatePayload(3.2);
// drone.updatePayload(98.6);
// drone.updatePayload(61.0);
// drone.updatePayload(19.3);
// drone.updatePayload(-67.0);
// drone.updatePayload(36.2);
// drone.updatePayload(-77.5);
// drone.updatePayload(3.6);
// drone.updatePayload(6.2);
// drone.updatePayload(17.8);
// drone.updatePayload(-15.5);
// drone.updatePayload(-62.1);
// drone.updatePayload(-33.7);
// drone.updatePayload(83.1);
// drone.updatePayload(30.8);
// drone.updatePayload(13.6);
// drone.updatePayload(-11.3);
// drone.updatePayload(61.3);
// drone.updatePayload(-92.8);
// drone.updatePayload(-24.3);
// drone.updatePayload(-47.8);
// drone.updatePayload(70.0);
// drone.updatePayload(8.1);
// drone.updatePayload(-27.5);
// drone.updatePayload(-96.8);
// drone.updatePayload(90.5);
// drone.updatePayload(53.9);
// drone.updatePayload(72.7);
// drone.updatePayload(5.2);
// drone.updatePayload(-30.2);
// drone.updatePayload(-27.8);
```

```
// drone.updatePayload(-15.6);
// drone.updatePayload(41.2);
// drone.updatePayload(-80.8);
// drone.updatePayload(51.5);
// drone.updatePayload(-74.8);
// drone.updatePayload(46.6);
// drone.updatePayload(-18.4);
// drone.updatePayload(-2.5);
// drone.updatePayload(35.2);
// drone.updatePayload(98.2);
// drone.updatePayload(9.4);
// drone.updatePayload(-43.2);
// drone.updatePayload(51.5);
// drone.updatePayload(-38.9);
// drone.updatePayload(44.0);
// drone.updatePayload(-83.4);
// drone.updatePayload(-87.3);
// drone.updatePayload(-14.3);
// drone.updatePayload(76.9);
// drone.updatePayload(-8.2);
// drone.updatePayload(-32.8);
// drone.updatePayload(-67.6);
// drone.updatePayload(-90.7);
// drone.updatePayload(92.6);
// drone.updatePayload(48.5);
// drone.updatePayload(84.5);
// drone.updatePayload(-32.2);
// drone.updatePayload(-3.6);
// drone.updatePayload(63.8);
// drone.updatePayload(-80.6);
// drone.updatePayload(75.0);
// drone.updatePayload(56.9);
// drone.updatePayload(26.7);
// drone.updatePayload(-25.2);
// drone.updatePayload(-78.4);
// drone.updatePayload(-63.6);
// drone.updatePayload(-5.5);
// drone.updatePayload(49.6);
// drone.updatePayload(-2.6);
// drone.updatePayload(63.4);
//testcase-FT:
// drone.updatePayload(-1.7976931348623141E308);
// drone.updatePayload(-1.4945527238855331E308);
// drone.updatePayload(0.0);
```

```
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(1.2478105289044308E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(4.093031884787999E290);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(7.022238808055921E305);
// drone.updatePayload(6.979279229465461E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(1.0219250716380363E303);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-5.780817139557251E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-0.0);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-8.389234629357474E307);
// drone.updatePayload(0.0);
// drone.updatePayload(6.979279229465461E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.6214487098758146E307);
// drone.updatePayload(7.543261389422266E307);
// drone.updatePayload(-1.7976931348623123E308);
// drone.updatePayload(7.022238808055921E305);
// drone.updatePayload(1.7976930712414922E308);
// drone.updatePayload(6.626813527751998E290);
// drone.updatePayload(1.6032990611555275E293);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(6.237000967295999E290);
// drone.updatePayload(3.995578744673999E290);
// drone.updatePayload(2.618542486109552E295);
// drone.updatePayload(5.173218082313993E295);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(6.237000967295999E290);
// drone.updatePayload(4.564455225236348E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(1.6859393239721996E291);
```

```
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(0.0);
// drone.updatePayload(0.0);
// drone.updatePayload(0.0);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(0.0);
// drone.updatePayload(-1.7204485079737006E308);
// drone.updatePayload(-1.7204485082872102E308);
// drone.updatePayload(6.415297069508656E307);
// drone.updatePayload(6.345074681428097E307);
// drone.updatePayload(6.345074681428097E307);
// drone.updatePayload(6.345074681428097E307);
// drone.updatePayload(6.345074681428097E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(6.237000967295999E290);
// drone.updatePayload(3.923073608429183E293);
// drone.updatePayload(1.0071821012037943E296);
// drone.updatePayload(6.600756214009271E300);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(7.022238808055921E305);
// drone.updatePayload(6.979279229465461E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(1.0219250716380363E303);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-5.780817139557251E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(7.022238808055921E305);
// drone.updatePayload(6.979279229465461E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(1.0219250716380363E303);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-5.780817139557251E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-1.7976931348623157E308);
```

```
// drone.updatePayload(7.022238808055921E305);
// drone.updatePayload(6.979279229465461E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(1.0219250716380363E303);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(-5.780817139557251E307);
// drone.updatePayload(-1.7976931348623157E308);
// drone.updatePayload(6.237000967295999E290);
// drone.updatePayload(4.564455225236348E307);
```

```
// testcase-Z3:
```

```
// drone.updatePayload(-55.0);
// drone.updatePayload(-71.9);
// drone.updatePayload(-41.7);
// drone.updatePayload(81.3);
// drone.updatePayload(-37.2);
// drone.updatePayload(94.2);
// drone.updatePayload(-59.9);
// drone.updatePayload(54.6);
// drone.updatePayload(77.8);
// drone.updatePayload(-78.5);
// drone.updatePayload(27.1);
// drone.updatePayload(37.1);
// drone.updatePayload(32.6);
// drone.updatePayload(-45.0);
// drone.updatePayload(-54.5);
// drone.updatePayload(-50.6);
// drone.updatePayload(1.2);
// drone.updatePayload(-58.2);
// drone.updatePayload(25.4);
// drone.updatePayload(39.7);
// drone.updatePayload(85.6);
// drone.updatePayload(-51.2);
// drone.updatePayload(-94.3);
// drone.updatePayload(-56.5);
// drone.updatePayload(-42.8);
// drone.updatePayload(20.5);
// drone.updatePayload(41.8);
// drone.updatePayload(30.4);
// drone.updatePayload(-66.9);
// drone.updatePayload(3.2);
// drone.updatePayload(98.6);
// drone.updatePayload(61.0);
// drone.updatePayload(19.3);
```

```
// drone.updatePayload(-67.0);
// drone.updatePayload(36.2);
// drone.updatePayload(-77.5);
// drone.updatePayload(3.6);
// drone.updatePayload(6.2);
// drone.updatePayload(17.8);
// drone.updatePayload(-15.5);
// drone.updatePayload(-62.1);
// drone.updatePayload(-33.7);
// drone.updatePayload(83.1);
// drone.updatePayload(30.8);
// drone.updatePayload(13.6);
// drone.updatePayload(-11.3);
// drone.updatePayload(61.3);
// drone.updatePayload(-92.8);
// drone.updatePayload(-24.3);
// drone.updatePayload(-47.8);
// drone.updatePayload(70.0);
// drone.updatePayload(8.1);
// drone.updatePayload(-27.5);
// drone.updatePayload(-96.8);
// drone.updatePayload(90.5);
// drone.updatePayload(53.9);
// drone.updatePayload(72.7);
// drone.updatePayload(5.2);
// drone.updatePayload(-30.2);
// drone.updatePayload(-27.8);
// drone.updatePayload(-15.6);
// drone.updatePayload(41.2);
// drone.updatePayload(-80.8);
// drone.updatePayload(51.5);
// drone.updatePayload(-74.8);
// drone.updatePayload(46.6);
// drone.updatePayload(-18.4);
// drone.updatePayload(-2.5);
// drone.updatePayload(35.2);
// drone.updatePayload(98.2);
// drone.updatePayload(9.4);
// drone.updatePayload(-43.2);
// drone.updatePayload(51.5);
// drone.updatePayload(-38.9);
// drone.updatePayload(44.0);
// drone.updatePayload(-83.4);
// drone.updatePayload(-87.3);
```

```
// drone.updatePayload(-14.3);
// drone.updatePayload(76.9);
// drone.updatePayload(-8.2);
// drone.updatePayload(-32.8);
// drone.updatePayload(-67.6);
// drone.updatePayload(-90.7);
// drone.updatePayload(92.6);
// drone.updatePayload(48.5);
// drone.updatePayload(84.5);
// drone.updatePayload(-32.2);
// drone.updatePayload(-3.6);
// drone.updatePayload(63.8);
// drone.updatePayload(-80.6);
// drone.updatePayload(75.0);
// drone.updatePayload(56.9);
// drone.updatePayload(26.7);
// drone.updatePayload(-25.2);
// drone.updatePayload(-78.4);
// drone.updatePayload(-63.6);
// drone.updatePayload(-5.5);
// drone.updatePayload(49.6);
// drone.updatePayload(-2.6);
// drone.updatePayload(63.4);
```

```
// testcase-UVT:
```

```
drone.updatePayload(-55.0);
drone.updatePayload(-71.9);
drone.updatePayload(-41.7);
drone.updatePayload(81.3);
drone.updatePayload(-37.2);
drone.updatePayload(94.2);
drone.updatePayload(-59.9);
drone.updatePayload(54.6);
drone.updatePayload(77.8);
drone.updatePayload(-78.5);
drone.updatePayload(27.1);
drone.updatePayload(37.1);
drone.updatePayload(32.6);
drone.updatePayload(-45.0);
drone.updatePayload(-54.5);
drone.updatePayload(-50.6);
drone.updatePayload(1.2);
drone.updatePayload(-58.2);
drone.updatePayload(25.4);
```

```
drone.updatePayload(39.7);
drone.updatePayload(85.6);
drone.updatePayload(-51.2);
drone.updatePayload(-94.3);
drone.updatePayload(-56.5);
drone.updatePayload(-42.8);
drone.updatePayload(20.5);
drone.updatePayload(41.8);
drone.updatePayload(30.4);
drone.updatePayload(-66.9);
drone.updatePayload(3.2);
drone.updatePayload(98.6);
drone.updatePayload(61.0);
drone.updatePayload(19.3);
drone.updatePayload(-67.0);
drone.updatePayload(36.2);
drone.updatePayload(-77.5);
drone.updatePayload(3.6);
drone.updatePayload(6.2);
drone.updatePayload(17.8);
drone.updatePayload(-15.5);
drone.updatePayload(-62.1);
drone.updatePayload(-33.7);
drone.updatePayload(83.1);
drone.updatePayload(30.8);
drone.updatePayload(13.6);
drone.updatePayload(-11.3);
drone.updatePayload(61.3);
drone.updatePayload(-92.8);
drone.updatePayload(-24.3);
drone.updatePayload(-47.8);
drone.updatePayload(70.0);
drone.updatePayload(8.1);
drone.updatePayload(-27.5);
drone.updatePayload(-96.8);
drone.updatePayload(90.5);
drone.updatePayload(53.9);
drone.updatePayload(72.7);
drone.updatePayload(5.2);
drone.updatePayload(-30.2);
drone.updatePayload(-27.8);
drone.updatePayload(-15.6);
drone.updatePayload(41.2);
drone.updatePayload(-80.8);
```

```

    drone.updatePayload(51.5);
    drone.updatePayload(-74.8);
    drone.updatePayload(46.6);
    drone.updatePayload(-18.4);
    drone.updatePayload(-2.5);
    drone.updatePayload(35.2);
    drone.updatePayload(98.2);
    drone.updatePayload(9.4);
    drone.updatePayload(-43.2);
    drone.updatePayload(51.5);
    drone.updatePayload(-38.9);
    drone.updatePayload(44.0);
    drone.updatePayload(-83.4);
    drone.updatePayload(-87.3);
    drone.updatePayload(-14.3);
    drone.updatePayload(76.9);
    drone.updatePayload(-8.2);
    drone.updatePayload(-32.8);
    drone.updatePayload(-67.6);
    drone.updatePayload(-90.7);
    drone.updatePayload(92.6);
    drone.updatePayload(48.5);
    drone.updatePayload(84.5);
    drone.updatePayload(-32.2);
    drone.updatePayload(-3.6);
    drone.updatePayload(63.8);
    drone.updatePayload(-80.6);
    drone.updatePayload(75.0);
    drone.updatePayload(56.9);
    drone.updatePayload(26.7);
    drone.updatePayload(-25.2);
    drone.updatePayload(-78.4);
    drone.updatePayload(-63.6);
    drone.updatePayload(-5.5);
    drone.updatePayload(49.6);
    drone.updatePayload(-2.6);
    drone.updatePayload(63.4);
}
}

```

*// only if currentPayload <= maxPayload, adjustFlightSpeed() will be called.*

*// hence, if (currentPayload <= maxPayload) in adjustFlightSpeed() will always be true.*

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 75%

| Element                        | Coverage | Covered Branches | Missed Branches | Total Branches |
|--------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool         | 0.4 %    | 3                | 839             | 842            |
| ▼ src                          | 0.4 %    | 3                | 839             | 842            |
| ▼ code                         | 0.4 %    | 3                | 839             | 842            |
| ▼ c86_SmartDroneDelivery.java  | 75.0 %   | 3                | 1               | 4              |
| ▼ c86_SmartDroneDelivery       | 75.0 %   | 3                | 1               | 4              |
| main(String[])                 |          | 0                | 0               | 0              |
| c86_SmartDroneDelivery(double) |          | 0                | 0               | 0              |
| adjustFlightSpeed()            | 50.0 %   | 1                | 1               | 2              |
| updatePayload(double)          | 100.0 %  | 2                | 0               | 2              |

3. Vibration testing instruction coverage: 98.5%

| Element                        | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|--------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ src                          | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ code                         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ c86_SmartDroneDelivery.java  | 98.5 %   | 385                  | 6                   | 391                |
| ▼ c86_SmartDroneDelivery       | 98.5 %   | 385                  | 6                   | 391                |
| main(String[])                 | 100.0 %  | 308                  | 0                   | 308                |
| c86_SmartDroneDelivery(double) | 100.0 %  | 17                   | 0                   | 17                 |
| adjustFlightSpeed()            | 100.0 %  | 23                   | 0                   | 23                 |
| updatePayload(double)          | 100.0 %  | 37                   | 0                   | 37                 |

4. Fuzz testing branch coverage: 75%

| Element                        | Coverage | Covered Branches | Missed Branches | Total Branches |
|--------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool         | 0.4 %    | 3                | 839             | 842            |
| ▼ src                          | 0.4 %    | 3                | 839             | 842            |
| ▼ code                         | 0.4 %    | 3                | 839             | 842            |
| ▼ c86_SmartDroneDelivery.java  | 75.0 %   | 3                | 1               | 4              |
| ▼ c86_SmartDroneDelivery       | 75.0 %   | 3                | 1               | 4              |
| main(String[])                 |          | 0                | 0               | 0              |
| c86_SmartDroneDelivery(double) |          | 0                | 0               | 0              |
| adjustFlightSpeed()            | 50.0 %   | 1                | 1               | 2              |
| updatePayload(double)          | 100.0 %  | 2                | 0               | 2              |

5. Fuzz testing instruction coverage: 98.5%

| Element                        | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|--------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ src                          | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ code                         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ c86_SmartDroneDelivery.java  | 98.5 %   | 385                  | 6                   | 391                |
| ▼ c86_SmartDroneDelivery       | 98.5 %   | 385                  | 6                   | 391                |
| main(String[])                 | 100.0 %  | 308                  | 0                   | 308                |
| c86_SmartDroneDelivery(double) | 100.0 %  | 17                   | 0                   | 17                 |
| adjustFlightSpeed()            | 100.0 %  | 23                   | 0                   | 23                 |
| updatePayload(double)          | 100.0 %  | 37                   | 0                   | 37                 |

6. Z3 branch coverage: 75%

| Element                        | Coverage | Covered Branches | Missed Branches | Total Branches |
|--------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool         | 0.4 %    | 3                | 839             | 842            |
| ▼ src                          | 0.4 %    | 3                | 839             | 842            |
| ▼ code                         | 0.4 %    | 3                | 839             | 842            |
| ▼ c86_SmartDroneDelivery.java  | 75.0 %   | 3                | 1               | 4              |
| ▼ c86_SmartDroneDelivery       | 75.0 %   | 3                | 1               | 4              |
| main(String[])                 |          | 0                | 0               | 0              |
| c86_SmartDroneDelivery(double) |          | 0                | 0               | 0              |
| adjustFlightSpeed()            | 50.0 %   | 1                | 1               | 2              |
| updatePayload(double)          | 100.0 %  | 2                | 0               | 2              |

7. Z3 instruction coverage: 98.5%

| Element                        | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|--------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ src                          | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ code                         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ c86_SmartDroneDelivery.java  | 98.5 %   | 385                  | 6                   | 391                |
| ▼ c86_SmartDroneDelivery       | 98.5 %   | 385                  | 6                   | 391                |
| main(String[])                 | 100.0 %  | 308                  | 0                   | 308                |
| c86_SmartDroneDelivery(double) | 100.0 %  | 17                   | 0                   | 17                 |
| adjustFlightSpeed()            | 100.0 %  | 23                   | 0                   | 23                 |
| updatePayload(double)          | 100.0 %  | 37                   | 0                   | 37                 |

## 8. Unimproved V-method branch coverage: 75%

| Element                        | Coverage | Covered Branches | Missed Branches | Total Branches |
|--------------------------------|----------|------------------|-----------------|----------------|
| ▼ VibrationTestingTool         | 0.4 %    | 3                | 839             | 842            |
| ▼ src                          | 0.4 %    | 3                | 839             | 842            |
| ▼ code                         | 0.4 %    | 3                | 839             | 842            |
| ▼ c86_SmartDroneDelivery.java  | 75.0 %   | 3                | 1               | 4              |
| ▼ c86_SmartDroneDelivery       | 75.0 %   | 3                | 1               | 4              |
| main(String[])                 |          | 0                | 0               | 0              |
| c86_SmartDroneDelivery(double) |          | 0                | 0               | 0              |
| adjustFlightSpeed()            | 50.0 %   | 1                | 1               | 2              |
| updatePayload(double)          | 100.0 %  | 2                | 0               | 2              |

## 9. Unimproved V-method instruction coverage: 98.5%

| Element                        | Coverage | Covered Instructions | Missed Instructions | Total Instructions |
|--------------------------------|----------|----------------------|---------------------|--------------------|
| ▼ VibrationTestingTool         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ src                          | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ code                         | 1.8 %    | 385                  | 21,134              | 21,519             |
| ▼ c86_SmartDroneDelivery.java  | 98.5 %   | 385                  | 6                   | 391                |
| ▼ c86_SmartDroneDelivery       | 98.5 %   | 385                  | 6                   | 391                |
| main(String[])                 | 100.0 %  | 308                  | 0                   | 308                |
| c86_SmartDroneDelivery(double) | 100.0 %  | 17                   | 0                   | 17                 |
| adjustFlightSpeed()            | 100.0 %  | 23                   | 0                   | 23                 |
| updatePayload(double)          | 100.0 %  | 37                   | 0                   | 37                 |